

Institute of Actuaries of India

Subject CM1A – Actuarial Mathematics (Paper A)

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:**i) Index linked security**

- Initial negative cash flow,
- followed by a series of unknown positive cash flows and a large final unknown positive cash flow, all on specified dates
- It is known that cash flows are linked to a certain index, such as inflation.
- Cash flows are known in “real terms”
- There is a lag in application of index to the cash flow amounts, due to delays in calculation of index
- It is also possible that borrower and lender agree to use index from an earlier period.

[3]

ii) Equity:

- Issued by commercial undertakings and other bodies.
- Entitle holders to receive a share in the profits of the company
- Distribution of profits to shareholders is in the form of dividends
- Dividends are paid after interest on loans and other fixed interest stocks has been paid.
- Higher expected returns than for most other asset classes
- There is a higher risk of capital losses, relative to fixed interest assets
- Returns on equity investments are not certain and can be variable
- Lowest ranking form of finance – in the event of company failure, shareholders will be entitled to any assets once all creditors have been paid
- Low initial running yield, dividends should increase with inflation
- Marketability varies according to size of company
- Voting rights in proportion to number of shares held.

[3]

[6 Marks]**Solution 2:****Advantages:**

- Allows to investigate the future behaviour of a process in the compressed time frame.
- Allows to study the stochastic nature of the results like Monte Carlo simulation.
- Helps in comparison of different scenarios to determine the best strategy.
- Allows to consider scenarios that would not be feasible in practice like because of cost or other business considerations.
- Allows control over experimental conditions, so that we can reduce the variance of the results output, without affecting the mean values

[2]

Disadvantages

- Might be time consuming.
- Might be expensive to set up.
- Stochastic models require large number of simulations to be carried out to get accurate results.
- More useful for comparing the results of input variations than for optimizing outputs.
- Give an impression of greater accuracy and reliability than they actually have, and so may create a sense of false security.
- The results from models are dependent on the data used, which may be inaccurate or unreliable.
- Users of the model may not understand fully how it works and its limitations.
- Models may not capture sufficiently accurately the real-world situation.
- Models require simplifying assumptions that might turn out to be wrong like they may ignore certain features (like inflation) or certain type of events that could occur (like catastrophes).
- Some models may be difficult to explain to clients.

[3]

[5 Marks]

Solution 3:

- i) Ultimate mortality is defined as the mortality which depends on age only [0.5]

Whereas select mortality depends on age and duration since some specific event [0.5]

The specific event might result from the policyholder having passed a medical test or have answered several questions about their health as part of the underwriting carried out by insurer before issuing a policy to the policyholder [1]

Therefore, the mortality of a policyholder aged say 35 with duration since inception of 1 year is expected to be better than a policyholder aged 35 years but duration since inception of 5 years as less time has elapsed since underwriting investigations for the former policyholder [1]
[3]

ii)

Age	Duration 0	Duration 1	Duration 2+ (Ultimate)
45	0.001201		
46	0.001320	0.001557	
47	0.001455	0.001735	0.001802
48		0.001938	0.002008
49			0.002241

Starting from l_{47} and working back upwards to $l_{[45]}$:

$$l_{[45]+1} = \frac{l_{47}}{(1 - q_{[45]+1})} = \frac{10000}{1 - .001557} = 10015.59 \quad [1]$$

$$l_{[45]} = \frac{l_{[45]+1}}{1 - q_{[45]}} = \frac{10015.59}{1 - .001201} = 10027.60 \quad [1]$$

Now working down from l_{47} to l_{48} :

$$l_{48} = l_{47} * (1 - q_{47}) = 10000(1 - .001802) = 9981.98$$

$$l_{[46]+1} = \frac{l_{48}}{1 - q_{[46]+1}} = \frac{9981.98}{1 - .001735} = 9999.33 \quad [1]$$

$$l_{[46]} = \frac{l_{[46]+1}}{1 - q_{[46]}} = \frac{9999.33}{1 - .001320} = 10012.55 \quad [1]$$

[4]
[7 Marks]

Solution 4:

$$96 = 4 a_{20|} + 100v^{20} \quad [1]$$

At $i=4\%$ RHS is 100 [0.5]

At $i=4.5\%$ $a_{20|}=13.00794$ $v^{20}=0.414643$; RHS=93.4961 [0.5]

Interpolation of $i=0.04+0.005 * ((100-96)/(100-93.4961))$
 $=4.3\%$ [1]
[3 Marks]

Solution 5:

Present value of benefit:

$$100,000A_{[45]:10}^1 = 100,000 (A_{[45]} - v^{10} {}_{10}p_{[45]}A_{55}) \quad [1]$$

$$= 100,000 (0.27583 - 1.04^{-10} \times \frac{9557.8179}{9798.0837} \times 0.3895) = 1915.02 \quad [1]$$

Variance:

$$= 100,000^2 \times (2A_{[45]:10}^1 - (A_{[45]:10}^1)^2) \quad [1]$$

$$= 100,000^2 \times \left({}^2A_{[45]} - v^{20} \frac{l_{55}}{l_{[45]}} {}^2A_{55} \right) - \left(\frac{1915.02}{100000} \right)^2 \quad [1]$$

$$= 100,000^2 \times \left((0.09431 - 1.04^{-20} \times \frac{9557.8179}{9798.0837} \times 0.17785) - 0.0191502^2 \right) = 147652397.80 \quad [1]$$

[5 Marks]

Solution 6:

i) Flat rate of interest

$$= (2 \times 2,400 - 2,000) / (2 \times 2,000) \quad [2]$$

$$= 70\% \quad [2]$$

[4]

ii) The flat rate of interest is not a good measure of the cost of borrowing because it takes no account of the timing of payments and the timing of repayment of capital [1]

iii) If the consumers' association is correct, then the present value of the repayments is greater than the loan at 200% [1]

$$\text{i.e. } 2000 < 2400 \cdot (i/d^{(12)}) \cdot a_{\overline{2}|} \quad [1]$$

$$i=2; a_{\overline{2}|}=0.44444; d^{(12)}=1.04982 \text{ gives RHS}=2032 \quad [1]$$

The consumers' association is correct. [1]

If the banks are correct, then the present value of the payments received by the bank, after expenses, is less than the amount of the loan at a nominal (before inflation) rate of interest of $(1.01463 \times 1.025 - 1)$ per annum effective = 0.04. [1]

$$\text{i.e. } 2000 > 720 \cdot (i/d^{(12)}) \cdot a_{\overline{2}|} + 720 \cdot (i/d^{(12)}) \cdot a_{\overline{1.5}|} + 960 \cdot (i/d^{(12)}) \cdot a_{\overline{1}|} - 0.3 \cdot 2400 \cdot (i/d^{(12)}) \cdot a_{\overline{2}|} \quad [1]$$

$$(i/d^{(12)}) = 1.021529; a_{\overline{2}|}=1.8861; a_{\overline{1}|}=0.9615; a_{\overline{1.5}|}=(1-1.04^{-1.5})/0.04=1.4283 \quad [1]$$

$$\text{So RHS} = 720 \times 1.021529 \times 1.8861 + 720 \times 1.021529 \times 1.4283 + 960 \times 1.021529 \times 0.9615 - 0.3 \times 2,400 \times 1.021529 \times 1.8861 \quad [1]$$

$$= 1,387.23 + 1,050.52 + 942.91 - 1,387.23 = 1,993.43 \quad [1]$$

Therefore, the banks are also correct [1]

[10]

[15 Marks]

Solution 7:**i) Decrement Table**

			[0.5]	[0.5]	[0.5]	[0.5]
Year	Age	lx	qx	No. of Deaths	No. of Maturity	No. of survivors
1	45	1.00000	0.001465	0.00147	-	0.99854
2	46	0.99854	0.001622	0.00162	-	0.99692
3	47	0.99692	0.001802	0.00180	0.99512	-

Unit Fund

	[0.5]	[1]	[0.5]	[1]		[0.5]	[0.5]
Year	Premium	Allocated Premium	Bid Offer Spread	Policy Fees	Fund after deduction	Fund after Growth	FMC
1	20,000.00	19,000.00	950.00	150.00	17,900.00	19,332.00	193.32
2	20,000.00	20,000.00	1,000.00	157.50	37,981.18	41,019.67	410.20
3	20,000.00	20,500.00	1,025.00	165.38	59,919.09	64,712.62	647.13

		[1]	[1]
Year	Fund after growth and FMC	Return of Policy Fees	Fund after return of Policy Fees
1	19,138.68	-	19,138.68
2	40,609.47	-	40,609.47
3	64,065.49	472.88	64,538.37

Non-Unit Fund

	[0.5]	[0.5]	[1]	[0.5]	[1]	[2]	
Year	Cost of Allocation	Bid Offer Spread	Policy Fees	Expense s	Commissio n	Non-Unit Investment Income	Death Benefit Cost
1	1,000.00	950.00	150.00	200.00	2,200.00	-18.00	87.90
2	-	1,000.00	157.50	100.00	600.00	27.45	97.18
3	-500.00	1,025.00	165.38	105.00	600.00	-0.88	115.09

	[0.5]	[1]	
Year	FMC	Return of Policy Fees	Profit
1	193.32	-	-212.58
2	410.20	-	797.97
3	647.13	472.88	43.66

[15]

- ii) The profit in non-unit fund in the first year is negative. This implies shareholder support is required in year 1 i.e. each policy sold will require the shareholder to inject money to support policy **[2]**

- iii) Risk Discount Rate is the rate of interest used to determine net present value of cash flows. **[1]**

The risk discount rate would typically be higher than the investment return assumption. **[1]**

Risk discount rate makes allowance for the risks associated with the business and would reflect the rate of return required by shareholders to take such risk **[1]**

[3]

- iv) In unit-linked products, there are usually two prices of each unit at the same time. Bid-price is the cash-in value of the unit and Offer price is the price that has to be paid to purchase a unit of the fund. The difference between the two is the bid-offer spread. [1]
- The offer price is typically higher than the bid-price. [0.5]
- Bid-offer spread is applied to each unit by a company and is a source of profit. [0.5]
- v) The profitability will increase. This is because the policy fee which is currently being repaid back to the policyholder, will now become a source of profit for the company [1]
- [23 Marks]**

Solution 8:

- i) The present value of 1st benefit is given by
- $$100000 * {}_5p_{55:50} * v^5$$
- $$= 100000 * (l_{60:55} / l_{55:50}) * v^5$$
- $$= 100000 * (l_{60} * l_{55} / (l_{55} * l_{50})) * v^5$$
- $$= 100000 * ((9826.131 * 9917.613) / (9904.805 * 9952.697)) * (1.04)^{-5}$$
- $$= 81252.42$$
- The present value of 2nd benefit is given by
- $$10000 * v^5 * ({}_5p_{55} * (\ddot{a}_{60} - 11/24) + {}_5p_{50} * (\ddot{a}_{55} - 11/24))$$
- $$= 10000 * 1.04^{-5} * ((9826.131 / 9904.805) * 15.174 + (9917.623 / 9952.697) * 17.752)$$
- $$= 269117.43$$
- The present value of 3rd benefit is given by
- $$100000 * (v_{10} * {}_{10}p_{55:50})$$
- $$= 100000 * (1.04)^{-10} * (1 - (1 - l_{65} / l_{55}) * (1 - l_{60} / l_{50}))$$
- $$= 100000 * 0.675564169 * (1 - (1 - 9647.797 / 9904.805) * (1 - 9848.431 / 9952.697))$$
- $$= 67538.05$$
- So, the present value of all the benefits is given by
- $$= 81252.42 + 269117.43 + 67538.05$$
- $$= 417907.90$$
- ii) The effective duration of the three possible asset classes are 5, 7 and 10 respectively. [1]
- The effective duration of the policyholder benefits should be greater than 5 years as significant benefits are paid after 5 years. Therefore, the 5 year ZCB would not be an appropriate choice. [1]
- The effective duration considering benefit 1 and 3 only will lie in the range of 5 – 10, say about 7. The 2nd benefit will increase the duration towards 10 or more. So, the ZCB with 10 year term may be most suited. [1]
- [3]**
- [13 Marks]**

Solution 9:

One year spot rate:

Zero coupon yield for year 1 = 4.5%, which is the GRY for the one year bond [0.5]

Two year spot rate:

Let the 2 year spot rate be i_2

$$5/(1+i_1) + 105/(1+i_2)^2 = 5a_{\overline{2}|5.3\%} + 100v^{2 \cdot 5.3\%} \quad [1]$$

$$5/(1.045) + 105/(1+i_2)^2 = 5(1-1/1.053^2)/0.053 + 100/1.053^2 \quad [1]$$

$$= 99.444539$$

$$(1+i_2)^2 = 105/94.659850$$

$$i_2 = 5.3202\% \text{ p.a.} \quad [1]$$

Three year spot rate:

The 3-year par yield is 5.6% p.a

$$1 = 0.056 \cdot \{1/(1+i_1) + 1/(1+i_2)^2 + 1/(1+i_3)^3\} + 1/(1+i_3)^3 \quad [1]$$

$$1.056/(1+i_3)^3 = 1 - 0.056/1.045 - 0.056/(1.053202)^2 \quad [1]$$

$$i_3 = 5.63242\% \text{ p.a.} \quad [1]$$

1-year forward rates:

$$f_0 = i_1 = 4.5\% \text{ p.a.} \quad [0.5]$$

$$(1+i_1) \cdot (1+f_1) = (1+i_2)^2$$

$$f_1 = 6.1468\% \text{ p.a.} \quad [1]$$

$$(1+i_2)^2 \cdot (1+f_2) = (1+i_3)^3$$

$$1+f_2 = (1.056324)^3 / (1.053202)^2$$

$$f_2 = 6.596\% \text{ p.a.} \quad [1]$$

[9 Marks]

Solution 10:

Let P be the monthly premium for the contract. Then

$$12P \ddot{a}_{[35]:25}^{(12)} = 12P \left[\ddot{a}_{[35]:25} - \frac{11}{24} (1 - v^{25} {}_{25}p_{[35]}) \right] \quad [1]$$

$$= 12P \left[16.029 - \frac{11}{24} \left(1 - 0.37512 \times \frac{9287.2164}{9892.9151} \right) \right] \quad [1]$$

$$= 188.7848P \quad [0.5]$$

EPV of death benefits:

$$260000 \bar{A}_{[35]:25}^1 - 10000 (IA)_{[35]:25}^1 = 10000 \times (1.04)^{0.5} \times [26A_{[35]:25}^1 - (IA)_{[35]:25}^1] \quad [1]$$

$$A_{[35]:25}^1 = A_{[35]:25} - v^{25} {}_{25}p_{[35]} = 0.38350 - 0.35220 = 0.03135 \quad [0.5]$$

$$(IA)_{[35]:25}^1 = (IA)_{[35]} - v^{25} {}_{25}p_{[35]} [25A_{60} + (IA)_{60}] \quad [1]$$

$$= 7.47005 - 0.35220 \times [25 \times 0.45640 - 8.36234] = 0.50722$$

$$10198.04(26 \times 0.03135 - 0.50722) = 3139.76 \quad [1]$$

EPV of annuity:

$$v^{25} {}_{25}p_{[35]} [47000 \ddot{a}_{60} + 3000 (I\ddot{a})_{60}]$$

$$= 0.35220 * [47000 \times 14.134 + 3000 \times 150.053] = 392456.13$$

[1]

[1]

EPV of expenses:

$$235 + 60 \left(\ddot{a}_{[35:25]}^{@0\%} - 1 \right) = 235 + 60 \left(e_{[35]} - \frac{l_{60}}{l_{[35]}} (1 + e_{60}) \right)$$

[2]

$$= 235 + 60 \left(43.909 - \frac{9287.2164}{9892.9151} \times 20.67 \right) = 1648.95$$

[1]

Initial and renewal commission expense:

$$0.030 \times 12P + 0.05 \times 12P \left(\ddot{a}_{[35:25]}^{(12)} - \frac{1}{12} \right) = 3.6P + 0.6P \times (15.732 - 0.08333)$$

$$= 12.98924P$$

[1]

$$188.7848P = 3139.76 + 392456.13 + 12.98924P + 1648.95$$

[1]

$$175.79558P = 397244.8$$

$$P = 2259.7$$

[1]

[14 Marks]
